Where will growth come from?

Reallocation, the missing factor in Hungary

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KRTK-KTI Working Paper, draft v1, 16 October 2026

Abstract

Hungary needs growth that does not burden the budget. We ask whether part of it can come from workers and capital moving from less to more productive firms. Using balance sheets of all Hungarian firms for 2004–2023 and CompNet data for Poland and Central and Eastern Europe, we run three exercises. By the late 2010s productivity gaps between firms in the same industry were wider in Hungary than in Poland and the region on every measure we have, and in Hungary they did not narrow after 2019. The contribution of resources moving between firms to productivity growth fell by almost half after 2013, from 1.6 to 0.8 percentage points a year. About 60% of this fall comes from foreign-owned firms; among domestic firms reallocation slowed by about a quarter. At the same time, the model's efficiency gap, the output lost to misallocation, widened among domestic firms at least as much as overall, most in construction and utilities. A standard model suggests that returning to the allocation of 2004–2010 would lift the level of output by 1–2% of GDP, a one-off but lasting gain; over five years, 0.2–0.4 percentage points of extra growth a year. Policy that gives firms more freedom could mobilise this reserve.

Keywords: productivity, reallocation, misallocation, firm dynamics, Hungary.
JEL: D24, L25, O47, O52.

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1 Introduction

Hungary runs a large budget deficit. For 2026 the Ministry of Finance expects 7.5% of GDP (Pénzügyminisztérium, 2026) and the central bank 6.9% (Magyar Nemzeti Bank, 2026), against the 3% limit of the Maastricht criteria. Bringing it down limits what the state can spend to support growth. This raises a practical question: are there resources already in the economy that could be mobilised by better economic policy, without new spending?

Growth comes from more workers, more capital and higher productivity. The first source is running out: employment grew by almost a fifth between 2010 and 2023, while the working-age population is shrinking.[1] The second needs investment that has to be financed, and Hungary already invests a lot: in 2018–2022 gross fixed capital formation was 25–28% of GDP a year, the third-highest rate in the EU after Ireland and Estonia, against 16–19% in Poland.[2] Hence growth has to come mostly from productivity, and a country that invests this much has reason to ask how well its capital is allocated.

Aggregate productivity rises in two ways. Firms can get better: they adopt new technology, improve management or find new markets. Or labour and capital can move from less productive to more productive firms in the same industry. This second channel, *reallocation*, requires markets in which good firms can grow and weak firms can shrink or close. The two channels are not separate from investment. New investment is itself a choice of which firms grow: capital that goes to productive firms raises aggregate productivity, capital that goes to weak or favoured ones does not. So the question is not only how much Hungary invests, but who invests, and where.

When rules, regulations or interventions make it harder for productive firms to expand or for weak firms to shrink, inputs stay where they produce less than they could. Economists call this *misallocation*. In the last decade Hungarian firms faced frequent changes in regulation, sector-specific taxes and, in 2021–2023, price caps on fuel, basic foods and mortgage rates. Each measure had its own goal. Taken together, they may also have changed which firms grow.

This paper asks how well the Hungarian economy allocates the workers and capital it already has, whether this has changed, and how much output a better allocation could add. We use the balance sheets of every Hungarian double-entry firm from 2004 to 2023 and run three exercises, each answering one question. The first asks whether resources are where they earn most, by comparing productivity gaps between firms with Poland and the region. The second asks how much growth still comes from resources moving to better firms, by splitting productivity growth into firms improving and resources moving. The third asks how much is at stake, using the model of Hsieh and Klenow (2009). All three compare firms within the same two-digit industry. Reallocation can also happen between industries, as workers move from manufacturing to services or from traditional to knowledge-intensive industries. We leave that channel aside: for Hungary, Muraközy et al. (2018) found a clear shift of employment from manufacturing to services until 2010 and nearly constant sector shares after, with the overwhelming majority of productivity

[1] Employed persons aged 20–64: 3.83 million in 2010, 4.55 million in 2023 (Eurostat `lfsi_emp_a`). Population aged 15–64: 6.87 million in 2010, 6.18 million in 2025 (Eurostat `demo_pjanbroad`).

[2] Eurostat `nama_10_gdp`, P51G as a share of GDP; average 2017–2022: Hungary 25.9%, EU 21.5%.

growth taking place within industries.

We find three things. First, by the late 2010s productivity gaps between firms in the same industry were wider in Hungary than in Poland and the region on every measure we have, and in Hungary they did not narrow after 2019. Second, the contribution of resources moving between firms to productivity growth fell by almost half after 2013, from 1.6 to 0.8 percentage points a year. About 60% of this fall comes from foreign-owned firms; among domestic firms reallocation slowed by about a quarter. Third, the model's efficiency gap rose by about six points of value added by 2016–2019 and by about ten points including 2020–2023, among domestic firms at least as much as overall, most in construction and utilities, and returns on capital became more uneven across firms. Returning to the allocation of 2004–2010 would lift the level of output by 1–2% of GDP, a one-off but lasting gain; over five years, 0.2–0.4 percentage points of extra growth a year.

The paper builds on earlier work on Hungary and the region. Muraközy et al. (2018) found strong reallocation before 2008 and much weaker reallocation after 2013, Koren and Rátfai (2026) find that the employment growth advantage of the most productive Hungarian firms has shrunk steadily since the 1990s, and CompNet data show that the countries that joined the EU in 2004 narrowed the productivity gaps between their firms towards the level of the old member states (Grassi, 2026). We contribute in three ways. First, we extend the evidence on dispersion and on the sources of productivity growth in Hungary to 2023. Second, we put Hungary next to Poland and the region on comparable measures. Third, we separate the contributions of foreign-owned and domestic firms.

The evidence is descriptive. We document how allocation changed, but we cannot tell which policy, if any, caused the change. Gaps in revenue productivity can reflect prices, markups and measurement as well as misallocation, and the model behind the third exercise is a benchmark, not an estimate of what a reform would deliver. Section 9.3 sets out which results are robust and which are not.

The paper proceeds as follows. Section 2 reviews the related literature and Section 3 describes the data. Sections 4, 5 and 6 present the three exercises, each with its method, results and robustness checks. Section 7 compares foreign-owned and domestic firms and Section 8 looks at four industries. Section 9 puts the results together, discusses policy and sets out the caveats, and Section 10 concludes. The appendix has supplementary figures and explains how the results can be reproduced.

2 Related literature

The paper draws on six groups of studies: the measurement of misallocation, the critiques of that measurement, the decline of reallocation in advanced economies, evidence from Europe and the region, work on Hungary, and studies that link specific policies to allocative efficiency.

Misallocation and the efficiency gap. Restuccia and Rogerson (2008) and Hsieh and Klenow (2009) started the modern literature. Restuccia and Rogerson show in a model that policies which tax some firms and subsidise others lower aggregate productivity even when no firm becomes less efficient. Hsieh and Klenow measure the dispersion of marginal revenue products across plants in China and India and compute how much output would rise if it were removed; this is the calculation of Section 6. Restuccia and Rogerson (2017) and Hopenhayn (2014) review the causes and the theory: taxes and regulations that depend on firm size or identity, financial frictions and weak property rights. Bartelsman et al. (2013) show that the covariance between firm size and productivity, a measure of allocative efficiency, differs widely across countries. The decomposition of Section 5 goes back to Baily et al. (1992) and Griliches and Regev (1995); we use the version of Foster et al. (2001) and check it with Melitz and Polanec (2015). Foster et al. (2008) show that revenue and physical productivity can differ, which matters because we observe revenue only.

What dispersion does and does not show. A wide spread of revenue productivity need not mean misallocation. Haltiwanger et al. (2018) show that it can also reflect differences in demand and technology that the standard model leaves out. In nine datasets covering 40 countries, Asker et al. (2014) find that an investment model with adjustment costs and volatile productivity accounts for 80–90% of the variation in the dispersion of returns to capital across industries and countries. David and Venkateswaran (2019) split the dispersion of returns to capital into its sources: in Chinese manufacturing, adjustment costs and uncertainty explain a modest share, and most comes from firm-specific factors correlated with productivity, the kind of wedge that policy can create. Bils et al. (2021) correct for measurement error in revenue and inputs; the correction lowers the estimated gain from reallocation by 20% in India and by 60% in the United States. With an EU-wide firm survey, Gorodnichenko et al. (2026) find that part of the measured dispersion reflects firm characteristics and may be compensation for differences across firms rather than a distortion. For this paper, these studies imply that we should read the level of the efficiency gap as a benchmark, not an estimate, and that its change over time is more informative than its level.

Declining reallocation and divergence. In the United States, Decker et al. (2017) link the productivity slowdown to weaker allocative efficiency, and Decker et al. (2020) show that firms have become less responsive to their own productivity: productive firms grow less than they used to. This is the closest parallel to the fall we find in Section 5 and to Koren and Rátfai (2026) for Hungary. In OECD data, Andrews et al. (2016) find that the productivity gap between frontier firms and the rest widened, more so in sectors with less extensive pro-competitive product market reforms, and Berlingieri et al. (2017) find that productivity dispersion rose within sectors in 16 countries, driven mainly by the bottom of the distribution (published as Berlingieri et al., 2024). Adalet McGowan et al. (2018) and Andrews et al. (2017) show that the capital sunk in zombie firms, old firms that persistently struggle to pay interest, rose from the mid-2000s, and that this goes with lower investment and employment growth of other firms in the industry and weaker reallocation of capital. Andrews and Petroulakis (2019) link zombie firms in Europe to weak

banks.

Europe and the region. CompNet, the source of our comparisons, is described by Lopez-Garcia et al. (2015) and CompNet (2021). Using firm data for five large euro-area countries in 2002–2012, Gamberoni et al. (2016) find that the allocation of capital worsened everywhere except Germany while the allocation of labour changed little, and that the deterioration was larger in services than in industry. This is the pattern we find in Section 6. Calligaris et al. (2018) estimate that Italy's aggregate productivity in 2013 would have been 18% higher had misallocation stayed at its 1995 level; the rise happened within sectors. Dias et al. (2016) find that misallocation within industries in Portugal almost doubled between 1996 and 2011, concentrated in services and on the capital side. Gopinath et al. (2017) link the rise in capital misallocation in Spain to the fall in interest rates after the introduction of the euro. For the region, Giordano and Lopez-Garcia (2018) use CompNet data for nine Central and Eastern European countries, Hungary and Poland among them, for 2003–2012, and find that more corruption goes with more misallocation of both labour and capital, especially where regulation is of lower quality. Grassi (2026) shows that the 2004 joiners narrowed the productivity gaps between their firms towards the level of the old member states, and Gradzewicz (2026) applies the Hsieh–Klenow calculation to Poland. At the level of the EU, the Draghi report attributes the productivity gap to the United States largely to the technology sector (Draghi, 2024); this paper looks at a different margin, allocation within industries.

Hungary. Muraközy et al. (2018) give the most detailed decomposition of Hungarian productivity growth before ours and find strong reallocation before 2008 and much weaker reallocation after 2013. Harasztosi (2011) decomposes Hungarian growth in 1994–2008 into capital, labour, productivity and reallocation. Koren and Rátfai (2026) find that the employment growth advantage of the most productive Hungarian firms over the median firm has shrunk steadily since the 1990s, and that the growth advantage of foreign-owned firms has disappeared. The latest OECD surveys reach a similar diagnosis. OECD (2024), citing Muraközy et al. (2018), reports that the within-firm, between-firm and entry and exit contributions to productivity growth all declined since the mid-2000s. OECD (2026) points to low business dynamism, product market regulation that holds back entry, insolvency rules that slow the reallocation of capital towards the most productive firms, and concentration and markups that rose faster than elsewhere in Europe. The central bank's productivity report documents a large gap between foreign-owned and domestic firms, about threefold in 2010–2016 and two-and-a-half-fold in 2021, but does not analyse allocation (Magyar Nemzeti Bank, 2024).

Several studies show how public resources were allocated across Hungarian firms. Exploiting a reform that allowed a more discretionary procedure below a contract-value threshold, Szucs (2024) finds that discretion in public procurement raised prices, selected less productive contractors and favoured firms connected to the governing party. Muraközy and Telegdy (2016) find that EU-funded grants in 2004–2012 favoured municipalities aligned with the governing coalition mainly for projects visible to voters, such as construction. Muraközy and Telegdy (2023) and

Banai et al. (2020) find that EU grants to small and medium-sized firms raised employment, sales and capital, but not clearly productivity. Varju and Papp (2022) analyse the sector-specific special taxes from a legal angle and argue that their design allowed discrimination, in particular against foreign-owned firms.

Policies and allocative efficiency. Andrews and Cingano (2014) link allocative efficiency to product market regulation, employment protection and barriers to entry, and Arnold et al. (2008) find that restrictive regulation of services goes with less efficient allocation within industries. Garicano et al. (2016) quantify one size-dependent rule: French labour regulations that bind at 50 employees act like a 2.3% tax on labour and cost 1.3–3.4% of GDP, depending on how flexible wages are. Midrigan and Xu (2014) find that financial frictions cost most through less entry and technology adoption, not through misallocation among incumbents. Fisman and Svensson (2007) find that bribery hurts firm growth more than taxation does. We found no study that links price caps to the allocation of resources across firms.

3 Data

3.1 Hungarian firms

Our main source is the Hungarian balance-sheet panel, Mérleg LTS, version 5.9.0 (November 2025). It covers every firm that keeps double-entry books. KRTK compiled it from corporate tax returns until 2014 and from Opten company records from 2015, with contributions by CEU MicroData. There is one record per firm and year, 1980–2023, 11.0 million firm-years; we use 2004–2023. The firm identifier (`frame_id`) follows firms through mergers and transformations; when it is missing we use the tax number. Industry codes are harmonised to two-digit NACE Rev. 2 and are year-specific.

For each firm and year we observe sales, material costs, value added, employment, labour cost, depreciation, tangible and intangible assets, equity, liabilities and the industry code. Value added is sales plus capitalised own work minus material costs (`gdp`); for the dispersion measure we use CompNet's definition, sales minus material costs. Labour is the number of employees (`emp`). Labour cost is payments to personnel including employer contributions (`persexp`). Capital is tangible plus intangible fixed assets at book value. Capital cost is depreciation (`kecs`), and in the version with imputed financing cost also the 10-year government bond yield (Eurostat `irt_lt_mcby_a`) times equity (`satok`, if positive) and long-term liabilities (`hlk`). Values are in thousand HUF, nominal unless stated.

3.2 Samples

Each exercise uses the sample its method and comparison call for. All three compare firms within the same two-digit industry; shifts of workers and capital between industries are not counted.

The first exercise follows CompNet: business entities (company-form codes 1xx) with at least 20 employees in CompNet's industries (NACE 10–33 without 19, 41–43, 45–47, 49–53, 55–56, 58–63, 68, 69–75, 77–82), with positive sales, value added, capital and materials, 2004–2023. This gives about 9,500–13,500 firms a year. CompNet's outlier routine is applied before the size cut.

The second and third exercises use the market economy (NACE 05–82 and 95, without 53 and 64–66), as in Muraközy et al. (2018); firms ever non-profit or budgetary are dropped. An active firm-year has at least one employee and positive value added and capital (the third exercise also requires a positive wage bill). We keep firms with at least five employees: 46,000–62,000 firms a year. For each window of the decomposition we keep firms with at least 5 employees in at least one end-year. Productivity is winsorised at the 1st and 99th percentiles by industry and year, and firms whose log productivity changes by more than 1.5 between consecutive 5+ years are dropped, as in Muraközy et al. (2018). Each three-year window has 34,000–58,000 continuing firms and 8,000–16,000 entrants and exiters.

3.3 Prices and coverage

A price index common to all firms in an industry and year does not change the first and third exercises, which compare firms within an industry and year: a deflator common to an industry-year cancels from the coefficient of variation and from the within-industry efficiency ratio. Differences in firms' own prices and markups do remain, because we observe revenue, not physical output. The decomposition measures growth over time and needs value added at constant prices. We use value-added price indices by A64 industry and the gross fixed capital formation price index from Eurostat (`nama_10_a64`, `nama_10_gdp`); air transport uses the transport total. As an alternative we use the producer price indices that are part of the panel. We show both, because they disagree in the inflation years 2021–2023.

The nominal value added of the decomposition sample is 62–64% of national-accounts gross value added in the same industries in every year 2015–2022 and grows at nearly the same rate (2022: 18.3% against 18.4%). In 2023 it grows by 5.9% against 15.9%. The number of active firms is stable (180,755 against 179,520), but this does not rule out changes in composition; the source of the gap is unknown, so we treat 2023 as provisional.

3.4 CompNet

For the comparison we use CompNet, which computes the same firm-level indicators with the same code in each member country (CompNet, 2021). We use its 8th vintage, firms with 20 or more employees, unweighted, two-digit industries, collapsed to country-year by Basile Grassi for Grassi (2026). It covers Hungary from 2004 and Poland from 2002, to 2019 (2018 for most old member states). The 9th vintage was examined and not used: its all-firms sample lacks Poland, and the fixed-weight residual is missing.

We compare Hungary with Poland and with the average of the Central and Eastern European

countries that joined the EU in 2004 and report to CompNet (the *CEE average*: the six 2004 joiners with data, Hungary included). The average of the old member states (nine EU-15 countries with data, shown when at least five report) serves as a reference level. CompNet's Hungarian data come from the same tax returns as ours, so for 2004–2019 we check our Hungarian series against CompNet's own (Section 4.3).

3.5 GDP and deficit

Nominal GDP in 2023 was HUF 75,218 billion (Eurostat `nama_10_gdp`, B1GQ, current prices). The general government balance was −7.0% of GDP in 2023, −5.1% in 2024 and −4.7% in 2025 (Eurostat `gov_10dd_edpt1`, April 2026 release). Forecasts for 2026 are −7.5% (Ministry of Finance, July 2026 budget review, the basis of the revised 2026 budget submitted in August; Pénzügyminisztérium, 2026) and −6.9% (Magyar Nemzeti Bank, 2026, September 2026 Inflation Report).

4 Productivity gaps within industries

The first exercise asks whether resources are where they earn most. If workers and capital move freely, productive firms expand and bid up the price of labour and capital until the extra return on their inputs falls to the level of their competitors, weak firms shrink, and the gaps between firms in an industry narrow over time. Because we measure revenue productivity, in a market without frictions it would be nearly equal across firms in an industry: a more productive firm sells more at a lower price until its revenue per unit of input matches its competitors'. A spread that stays wide suggests that resources are not moving to where they earn most (Hsieh and Klenow, 2009), though measurement and market power also widen it.

4.1 Measure

For every firm we compute total factor productivity (TFP) as a Solow residual: what is left of value added after labour and capital, each weighted by its share in costs, are accounted for. Because we observe sales and not physical output, it is *revenue productivity*: a firm that charges higher prices looks more productive. For firm i in industry s and year t,

$$SR_{ist} = \frac{VA_{ist}}{L_{ist}^{\alpha_{st}} \, K_{ist}^{1-\alpha_{st}}}, \tag{1}$$

where α_{st} is the median across firms in the industry and year of labour cost divided by labour cost plus capital cost. Constant returns. Baseline: capital cost = depreciation. Second version: depreciation + r_t (equity + long-term debt) with r_t the nominal 10-year bond yield, an accounting proxy for CompNet's capital cost, which includes interest paid.

We drop outliers with CompNet's routine (CompNet, 2021): a firm-year is dropped if any of six ratios (sales to labour, labour cost, capital and materials; capital to labour; materials to labour)

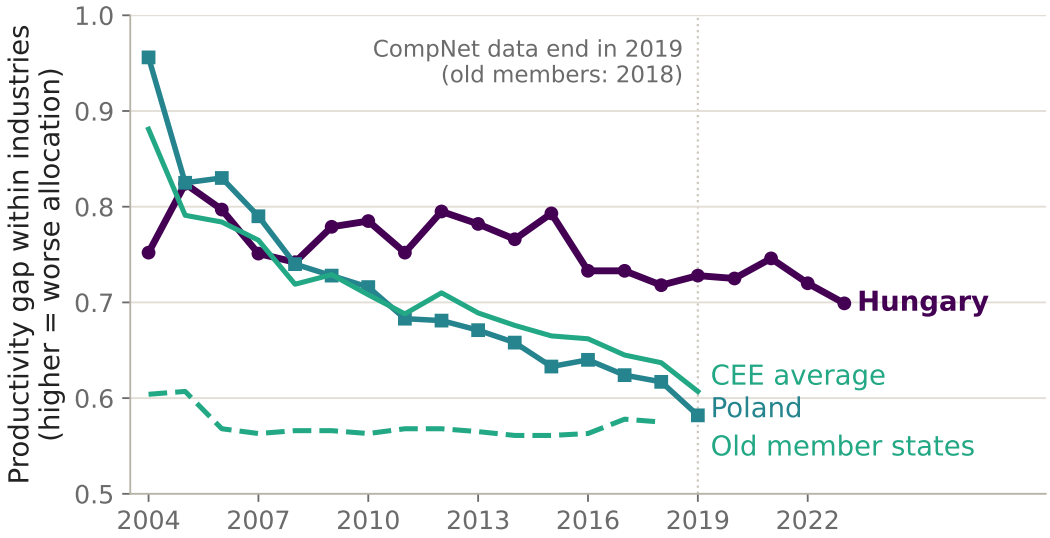


Figure 1: Productivity gaps between firms in the same industry, 2004–2023

Notes: Standard deviation divided by the mean of firm-level TFP (Solow residual with time-varying cost-share weights, cost of capital equal to depreciation) within two-digit industries, combined across industries as a root mean square weighted by the number of firms; firms with 20 or more employees, CompNet industries. Hungary: own calculation from the Mérleg LTS panel. Poland, CEE average (2004 EU joiners reporting to CompNet; Hungary is part of this average) and old member states (9 EU-15 countries with data, shown while at least 5 report): CompNet 8th vintage, aggregated by B. Grassi. A version with an imputed financing cost in the cost of capital is in Figure 2.

is in the bottom 3% or top 2% of its industry-year distribution, computed on all firms before the 20-employee cut. This reproduces CompNet's firm counts within 2.5% in 2004–2015.

For each industry-year with at least 10 firms, we take the standard deviation divided by the mean, $CV_{st} = \mathrm{sd}(SR)/\mathrm{mean}(SR)$, and combine industries into one number, giving more weight to industries with more firms:

$$CV_t = \Big(\sum_s N_{st}\, CV_{st}^2 \Big/ \sum_s N_{st} \Big)^{1/2},$$

with N_{st} the number of firms (Grassi's aggregation). Regional averages apply the same formula across countries. This is CompNet's measure, so Hungary can be put next to Poland and the region. For Hungary we use our own calculation, 2004–2023; for Poland, the CEE average and the old member states, CompNet's series computed the same way, to 2019 (CompNet, 2021; Grassi, 2026).

4.2 Results

Figure 1 shows the spread of TFP between firms in the same industry, combined over industries; higher values mean wider gaps. In the mid-2000s Hungary, Poland and the region were at about the same level, 0.76–0.80, well above the old member states at 0.58. Poland's line then fell almost every year and by 2019 reached the level of the old member states. Hungary's rose during the crisis, fell back in 2016, the year after the data source changed (CompNet's own Hungarian series falls at the same time), and stayed around 0.72–0.75.

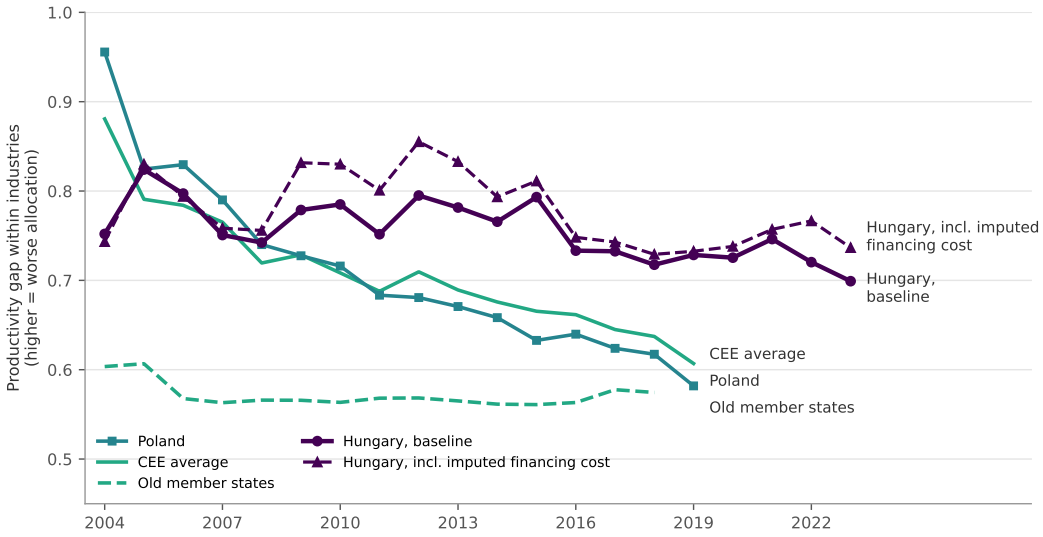


Figure 2: Productivity gaps with an imputed financing cost in the cost of capital, 2004–2023
Notes: As Figure 1, with a second Hungarian line in which the cost of capital also includes the 10-year government bond yield times equity and long-term debt. The two Hungarian lines differ by 0.04–0.06 in 2009–2013 and 2022–2023 and are close otherwise.

In period averages, our Hungarian series is 0.78 in 2005–2008, 0.73 in 2016–2019 and 0.70 in 2023; Poland's is 0.80 and 0.62, the CEE average 0.76 and 0.64, and the old member states 0.58 and 0.57. In 2019 the Hungarian value was 0.15 above the Polish one (CompNet's own Hungarian line: 0.69 against 0.58). So by 2019 the Hungarian gap was clearly larger than the Polish one, and CompNet's own Hungarian series gives the same picture. Adding an imputed financing cost to the cost of capital raises the Hungarian line somewhat when interest rates were high, in 2009–2013 and 2022–2023, by 0.04–0.06, and changes nothing else (Figure 2).

Persistent gaps mean that more productive Hungarian firms did not grow enough to close them, or less productive ones did not shrink. This is consistent with an economic environment that held resources in less productive firms. It is also consistent with other explanations, such as a different industry mix, differences in prices and markups, or measurement. This exercise cannot tell these apart.

4.3 Validation against CompNet

CompNet reports mean log labour productivity $E[\ln(VA/L)]$ and mean log Solow residuals with fixed (2/3) and cost-share weights. Since $\ln SR = \ln(VA/L) - (1-\alpha)\ln(K/L)$, we can back out an implied labour weight:

$$E[\ln(K/L)] = 3\left(E[\ln(VA/L)] - E[\ln SR^{2/3}]\right), \qquad \bar{\alpha} = 1 - \frac{E[\ln(VA/L)] - E[\ln SR^{cs}]}{E[\ln(K/L)]}. \tag{2}$$

With industry-specific weights $\bar{\alpha}$ is a composite, $E[\alpha_s \ln(K/L)]/E[\ln(K/L)]$, not the average labour share, and it depends on the units of K/L. We use it as a diagnostic only. For Hungary it rises from 0.75 (2004) to 0.84 (2019), for Poland from 0.76 to 0.86. On our data the composite is 0.004–0.007 below the mean weight.

Table 1: Our Hungarian series against CompNet's, 2004–2019

Labour weight	CompNet series	Mean gap	RMSE	Correlation	Labour-weight gap
Fixed weights, labour 2/3	PV08	−0.009	0.026	0.83	–
LC / VA	PV09	0.084	0.100	0.16	−0.10 [−0.18, −0.05]
LC / (LC + depreciation)	PV09	0.002	0.021	0.86	0.07 [0.03, 0.09]
LC / (LC + depr. + interest)	PV09	0.024	0.037	0.75	−0.02 [−0.04, −0.01]

Notes: Gap = our value minus CompNet's. PV08: CompNet's Solow residual with fixed weights; PV09: with cost shares. Labour-weight gap: our implied composite coefficient minus CompNet's, mean and [range]. Number of firms within 2.5% of CompNet's in every year 2004–2015.

Table 2: Three CompNet measures of productivity gaps, period averages

	A. TFP, cost-share weights			B. TFP, fixed weights			C. Labour productivity		
	2005–08	2016–19	Change	2005–08	2016–19	Change	2005–08	2016–19	Change
Hungary	0.78	0.71	−0.08	0.83	0.83	0.00	0.85	0.73	−0.12
Poland	0.80	0.62	−0.18	0.73	0.73	0.00	0.74	0.62	−0.12
CEE average (2004 EU joiners)	0.76	0.64	−0.13	0.76	0.75	−0.01	0.76	0.65	−0.11
Old member states (EU-15)	0.58	0.57	0.00	0.77	0.70	−0.06	0.59	0.60	0.01

Notes: Standard deviation divided by the mean within two-digit industries; CompNet 8th vintage, firms with 20 or more employees, all from CompNet's own series (Hungary is CompNet's Hungarian line, slightly different from ours). Old member states: 2016–2018. Fixed weights: labour 2/3, capital 1/3. Changes are computed from unrounded values.

Table 1 shows the check. With fixed weights our series matches CompNet's closely (mean gap −0.009), which validates the sample and data. With cost shares, labour cost over value added fails (correlation 0.16); the two versions we report are within 0.002 (baseline) and 0.024 (imputed financing cost) of CompNet's on average. After 2015 our firm count grows while CompNet's stays flat, and our line is slightly above CompNet's (2019: 0.73 against 0.69); we have not established why.

4.4 Three productivity measures

Productivity can be measured in more than one way, and the weights given to labour and capital are a choice. Let us check whether the comparison survives this choice. We compute the same spread with three productivity measures that CompNet publishes for the same firms: labour productivity (value added per employee), TFP with fixed weights (two thirds labour, one third capital), and TFP with cost-share weights, the measure of Figure 1. We use CompNet's own series for Hungary, Poland, the CEE average and the old member states, and compare the averages of 2005–2008 and 2016–2019 (CompNet, 2021); Figure A1 in the appendix shows the yearly series.

Table 2 gives the levels. In 2016–2019 Hungary's gaps were wider than Poland's on all three measures, by 0.09–0.11, and wider than the CEE average by 0.07–0.08. Poland's gaps were close to those of the old member states on all three. This level difference does not depend on how productivity is measured.

How the difference arose does depend on the measure. On the cost-share measure Hungary and Poland started at about the same level in 2005–2008, and Poland's gaps narrowed much more. On labour productivity Hungary's gaps were already about 0.1 wider in 2005–2008, and both countries narrowed by the same 0.12. With fixed weights neither country changed. So we

Table 3: Productivity gaps in Hungary on three measures, our own data, 2005–2008 to 2020–2023

	2005–08	2016–19	2020–23	Change to 2016–19	Change to 2020–23
TFP, cost-share weights (Exercise 1)	0.78	0.73	0.72	−0.05	−0.01
TFP, fixed weights (labour 2/3)	0.81	0.81	0.80	0.00	0.00
Labour productivity	0.84	0.76	0.75	−0.08	−0.01

Notes: Standard deviation divided by the mean within two-digit industries, combined as in Figure 1; firms with 20 or more employees in CompNet's industries, own calculation from the Mérleg LTS panel. Cost-share weights: labour cost over labour cost plus depreciation, the Hungarian line of Figure 1. CompNet data, and hence Poland, end in 2019. With 2021–2023 instead of 2020–2023 the values are the same to two decimals. Changes are computed from unrounded values.

do not read the comparison as Hungary falling behind Poland. Nor do we take a view on whether the difference lies with labour or with capital: the measures weight the two differently, and the results move with the weights.[3]

CompNet's data end in 2019, so Poland cannot be followed further. For Hungary we can extend all three measures with our own data (Table 3); our labour-productivity series follows CompNet's closely in 2004–2019 (mean gap 0.002, correlation 0.91). Between 2016–2019 and 2020–2023 the gaps hardly changed: from 0.73 to 0.72 on the cost-share measure, from 0.81 to 0.80 with fixed weights and from 0.76 to 0.75 on labour productivity. The narrowing on labour productivity before 2019 did not continue. This has two implications. First, the gap to Poland and the old member states that Figure 1 shows for 2019 did not close afterwards on the Hungarian side; we cannot see whether Poland's gaps moved. Second, the shocks of 2020–2023 (the pandemic, the energy shock and the price caps) could have pushed resources towards better firms or kept them in place. On these measures the gaps neither narrowed nor widened, so by this yardstick allocation did not improve.

Taken together, the three measures agree that Hungarian firms in the same industry are further apart than Polish firms and firms in the region, and that in Hungary these gaps did not narrow after 2019. They do not agree on when the difference opened, or on whether it lies with labour or with capital. A decomposition of the cost-share measure into labour-productivity dispersion, capital-intensity dispersion and their covariance under frozen weights would separate data changes from weight changes; it can be done on Hungarian data, not on CompNet aggregates.

5 Sources of productivity growth

The first exercise describes the state of allocation in a given year. The second looks at the flow: how much productivity growth comes from workers moving between firms and from firms entering and leaving. An industry's productivity can grow even if no firm gets better: when employment shifts from weaker to stronger firms, when weak firms leave and when strong ones enter. If this part shrinks, one of the two channels of productivity growth weakens. We first split growth into its parts, then follow the part that comes from resources moving over two decades. Section 7 asks

[3]The cost-share measure weights capital by its share in costs, and as interest rates fell the labour weight rose from about 0.75 to 0.85 in both countries. Changes in this measure therefore mix changes in the data with changes in the weights; CompNet's aggregates cannot separate the two (Section 4.3).

how much of its change comes from foreign-owned firms.

5.1 Decomposition

Aggregate productivity in an industry is the employment-weighted average of its firms' productivity. Following Foster et al. (2001), the change in an industry's productivity over three years splits into four parts: (i) *within firms*, as continuing firms improve; (ii) *reallocation*, as employment shifts between continuing firms towards more productive ones; (iii) *entry*; and (iv) *exit*. The second, third and fourth parts together are the growth that comes from resources moving. The split follows employment, so it shows where workers go; TFP itself includes capital.

Firm productivity is $tfp_{it} = \ln VA_{it} - \alpha_s \ln L_{it} - (1-\alpha_s) \ln K_{it}$, with α_s the median over 2004–2023 of labour cost divided by labour cost plus depreciation plus the imputed financing cost, among 5+ firms in industry s (median 0.78, range 0.17–0.97). The weight is fixed over time. Value added is deflated by the industry value-added price index (Eurostat, 2026), capital by the gross fixed capital formation deflator; the producer-price version deflates both by the industry PPI.

Between years a and b, in industry s, with employment shares θ_{it} and $P_t = \sum_i \theta_{it}\, tfp_{it}$,

$$
\begin{aligned}
P_b - P_a = & \sum_{i \in C} \theta_{ia}\, \Delta tfp_i && \text{(within)} \\
& + \sum_{i \in C} (tfp_{ia} - P_a)\, \Delta\theta_i && \text{(between)} \\
& + \sum_{i \in C} \Delta tfp_i\, \Delta\theta_i && \text{(cross)} \\
& + \sum_{i \in N} \theta_{ib}\, (tfp_{ib} - P_a) && \text{(entry)} \\
& - \sum_{i \in X} \theta_{ia}\, (tfp_{ia} - P_a) && \text{(exit)} \qquad (3)
\end{aligned}
$$

C, N and X are continuing, entering and exiting firms *of industry* s. The terms add up exactly (asserted in the code). We decompose within each two-digit industry and aggregate industry terms with the average of the industry's employment share at a and b. Windows are 2004–07, 2007–10, 2010–13, 2013–16, 2016–19, 2019–22 and 2022–23; values are divided by window length, and period values are averages weighted by length.

Entry and exit mean entering or leaving the observed sample of an industry. Most are firms appearing in or disappearing from the data; some are continuing firms that changed industry, and some disappearances are firms that stopped reporting or fell below the size rule, not closures. A firm active at both ends but in different industries is an exiter from one and an entrant into the other: 765–5,578 firms per window, 17–41% of entrants and 17–35% of exiters.

The total is the change in the employment-weighted average of firms' log TFP within industries. It is not aggregate TFP from the national accounts, which weights firms by output and covers the whole economy, and it can grow much faster, as in 2016–2019. We use it only to split growth into its sources. We use employment shares rather than value-added shares: with value-added TFP, value-added shares and the productivity measure share the same measurement error, which

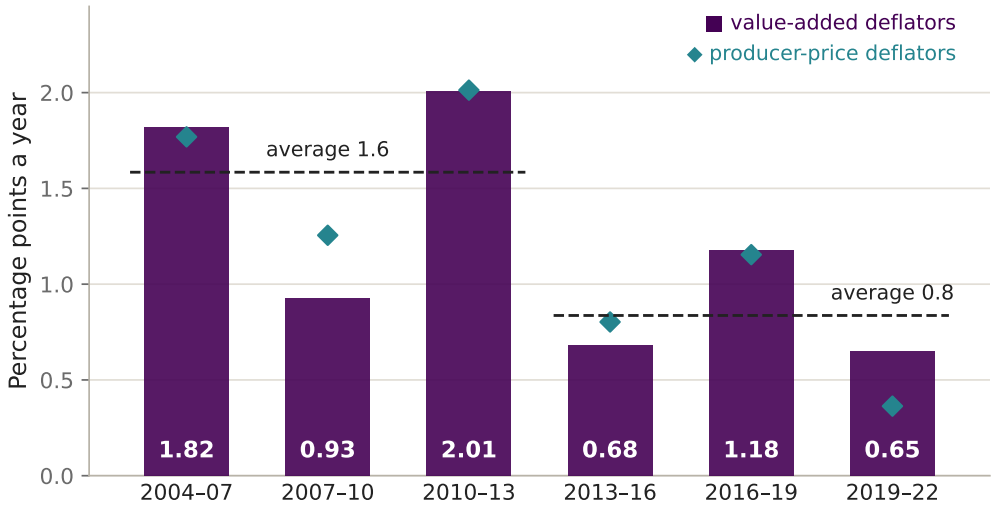


Figure 3: Contribution of resources moving between firms to productivity growth, by three-year window
Notes: Reallocation (between and cross terms) plus net entry and exit, annual percentage points; Foster–Haltiwanger–Krizan decomposition within two-digit industries, employment weights, market economy, firms with 5+ employees. Bars: value-added deflators; diamonds: producer price indices. Dashed lines: averages for 2004–2013 and 2013–2022. The provisional one-year window 2022–2023 is not shown. Period averages of all components are in Table 4.

inflates the cross term (3.5 points a year in 2004–2010 in our data).

Productivity contains $-\alpha_s \ln L$, so a firm that adds workers before output follows, or whose employment is mismeasured, shows a negative co-movement of productivity and share. A negative cross term is therefore consistent with resources not following productivity gains, but also with adjustment lags and measurement error. As a sensitivity we split it equally: within + cross/2 and between + cross/2.

We use all market-economy firms with at least five employees, 2004–2022, in three-year windows; 2022–2023 is reported in Table 4 but treated as provisional.

5.2 Results

Figure 3 shows how many percentage points a year resources moving added to productivity growth in each three-year window. The series is volatile: 1.82 points a year in 2004–2007, 0.93 in 2007–2010, 2.01 in 2010–2013, 0.68 in 2013–2016, 1.18 in 2016–2019 and 0.65 in 2019–2022. Part of these swings follows the business cycle and macroeconomic shocks, such as the crisis of 2008–2009 and the years 2020–2022, so a single window says little. We therefore compare averages over long periods. Averaged before and after 2013, the contribution fell by almost half, from 1.6 points a year in 2004–2013 to 0.8 in 2013–2022 (1.7 and 0.8 with producer prices). Every window after 2013 is below the earlier average. The two weakest are 2013–2016 and 2019–2022, the years of the pandemic, the energy shock and the price caps.

Table 4 shows all components before and after 2013 and in the later windows. Growth itself did not collapse: firms improved much faster after 2013 (2.8 points a year against 0.6), so productivity growth since 2013 came mostly from firms improving themselves. As the last rows of the table

Table 4: Sources of productivity growth, annual percentage points

	2004–13	2013–22	2013–16	2016–19	2019–22	2013–23
Firms improving (within)	0.55	2.84	1.85	3.95	2.71	1.76
Between	1.12	0.84	0.65	0.80	1.06	0.85
Cross	−0.53	−0.60	−0.44	−0.41	−0.95	−0.68
Reallocation (between + cross)	0.59	0.23	0.21	0.39	0.10	0.17
Entry	−0.31	0.00	−0.34	0.44	−0.10	−0.10
Exit	1.31	0.60	0.81	0.35	0.65	0.60
Net entry and exit	1.00	0.60	0.47	0.79	0.55	0.50
Resources moving (reallocation + net entry)	1.58	0.84	0.68	1.18	0.65	0.67
Total	2.14	3.67	2.53	5.13	3.36	2.43
Share of total growth from resources moving	74%	23%	27%	23%	19%	28%
Resources moving, producer prices	1.68	0.77	0.80	1.15	0.36	0.66
Share of total growth, producer prices	72%	26%	24%	23%	102%	26%

Notes: Foster–Haltiwanger–Krizan decomposition within two-digit industries, aggregated with industry employment weights; annualised log changes ×100. Three-year windows; 2004–13 and 2013–22 average three windows each; 2013–16, 2016–19 and 2019–22 are single windows; 2013–23 adds the one-year window 2022–23, which is provisional. Value-added deflators unless stated. Shares of total growth are not meaningful when total growth is close to zero (producer prices, 2019–22: total 0.35). Firms with at least 5 employees in at least one end-year; market economy.

show, the share of growth that came from resources moving fell from three-quarters (74%) in 2004–2013 to under a quarter (23%) in 2013–2022. This share falls more than the contribution itself, because the total grew.

Three details help to read the numbers. First, firms that leave are less productive than average, and in most windows entrants are not more productive than incumbents, so the gain from entry and exit comes from exit; the exception is 2016–2019, when entry added more (0.44 points) than exit (0.35). Second, between 2004–2013 and 2013–2022 the contribution of resources moving fell by about 0.75 points a year. About half of the fall comes from net entry and exit: weak firms leaving added much less (1.3 points a year before 2013, 0.6 after), only partly offset by entrants that pulled productivity down less. The other half comes from reallocation among continuing firms, mostly from the part that moves employment towards initially productive firms (the between term, 1.1 points a year before 2013 and 0.8 after). Third, the cross term, firms whose TFP rose gaining fewer workers, is about the same on average before and after 2013, but it is most negative in 2019–2022 (−0.95).

The weaker between term means that employment shifted less towards the firms that were already more productive. A negative cross term can mean that workers do not follow productivity gains, but it can also reflect adjustment lags, labour hoarding in a crisis, or measurement error, because employment enters both productivity and the shares. If the cross term is split equally between firms improving and reallocation, the fall in resources moving is about the same size. It fits Koren and Rátfai (2026), who find that the growth advantage of the most productive Hungarian firms has shrunk.[4] Whether the weakening of 2019–2022 persists once the crisis years

[4]We take snapshots, Koren and Rátfai (2026) look at change. They ask whether firms with higher value added per worker than others in their industry grow faster over the next three years, and find that this growth advantage has shrunk. Sections 4 and 6 measure how unequal productivity is in a given year. The two are sides of the same mechanism. In a market that works, firms with high revenue productivity grow, and as they grow their revenue per unit of input falls back towards that of their competitors, so the gaps narrow. When productive firms no longer outgrow the

Table 5: Robustness of the decomposition: resources moving (reallocation + net entry and exit), annual percentage points

	04–13	13–22	13–16	16–19	19–22	13–23
Main: within industries, value-added deflators	1.58	0.84	0.68	1.18	0.65	0.67
Producer price index deflators	1.68	0.77	0.80	1.15	0.36	0.66
Labour weight: LC / (LC + depreciation)	1.54	0.80	0.57	1.21	0.63	0.61
Entry = crossing 5 employees (MRB sample)	1.81	1.07	0.73	1.48	1.00	0.89
Firms changing industry dropped	1.80	0.79	0.72	1.06	0.60	0.64
Melitz–Polanec (2015)	2.02	−0.10	0.47	0.37	−1.15	−0.19
Main, cross term split equally (between + cross/2 + net entry)	1.85	1.14	0.90	1.38	1.13	1.01

Total productivity growth, same specifications

	04–13	13–22	13–16	16–19	19–22	13–23
Main: within industries, value-added deflators	2.14	3.67	2.53	5.13	3.36	2.43
Producer price index deflators	2.34	2.95	3.41	5.09	0.35	2.55
Labour weight: LC / (LC + depreciation)	2.22	3.65	2.65	5.38	2.92	2.41
Entry = crossing 5 employees (MRB sample)	2.20	3.67	2.51	5.14	3.37	2.44
Firms changing industry dropped	2.29	3.72	2.64	5.15	3.36	2.47
Melitz–Polanec (2015)	2.14	3.67	2.53	5.13	3.36	2.43

Notes: 04–13 and 13–22 average three windows each; 13–16, 16–19 and 19–22 are single windows; 13–23 adds the one-year 2022–23 window. Labour weight: without the financing cost. MRB sample: firm-years with 5+ employees, so crossing the line counts as entry or exit. Firms changing industry dropped: such firms removed from both ends of each window. Melitz–Polanec: reallocation = change in the size–productivity covariance among survivors.

pass is an open question; the weak window of 2013–2016 was followed by a recovery.

5.3 Robustness

These choices affect the levels, not the main finding. Table 5 shows what happens when we change them: producer price indices instead of value-added deflators, a labour weight without the financing cost, entry and exit defined by crossing the five-employee line (the MRB sample), dropping firms that change industry, splitting the cross term equally between firms improving and reallocation, and the decomposition of Melitz and Polanec (2015). Figure 4 shows the reallocation term alone.

The contribution of resources moving is lower in 2013–2022 than in 2004–2013 in every specification, and lower in 2019–2022 than in 2016–2019 in every one. Dropping firms that change industry raises the early contribution (1.80 in 2004–2013) and leaves the later one nearly unchanged (0.79). The equal split of the cross term leaves the decline about the same size (1.85 to 1.14). The Melitz–Polanec decomposition makes the decline larger. Total growth depends on the price index after 2019: with producer prices the 2019–2022 window grows 0.4% a year, with value-added prices 3.4%. Figure A2 in the appendix shows all components with both price indices.

The one-year window 2022–2023 is provisional. With value-added deflators it has within-firm growth of −7.9 points and a total of −8.8; with producer prices −0.7 and −1.1. The national-accounts price of value added rose by 16–19% in 2023 while our firms' nominal value added grew

rest, the gaps that new shocks create stay open. The between term of the decomposition, employment shifting towards initially more productive firms, is the aggregate counterpart of their growth advantage, and it also fell after 2013.

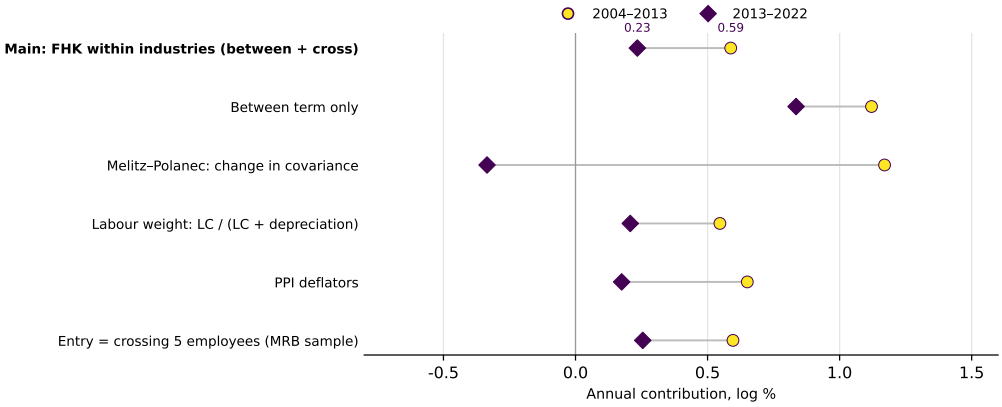


Figure 4: Reallocation by method, before and after 2013

Notes: The reallocation term (between + cross; Melitz–Polanec: change in covariance) by specification, annual percentage points, averages of the three-year windows in 2004–2013 and 2013–2022. The figure also shows the between term alone, which is not in Table 5.

by 6%. Reallocation and exit are nearly the same under both deflators, so the problem affects the within term and the total, not the finding about reallocation.

5.4 Comparison with Muraközy et al. (2018)

Their Figure 6.3 shows the same decomposition for 2004–2016. Our version differs in four ways. (i) We decompose within industries: with industry-specific weights, TFP levels of different industries are on different scales. Measuring capital in million instead of thousand forints shifts each firm's TFP by $(1 - \alpha_s) \ln 1000$; the within-industry results do not move, while the pooled 2004–2010 reallocation term moves from 1.50 to 0.97. Hence we decompose within industries, where the result does not depend on units. (ii) Entry and exit are defined by presence in an industry's sample, not by crossing the 5-employee line. (iii) Cost-share weights rather than estimated production functions (Ackerberg et al., 2015). (iv) Data to 2023 and two price indices. With their sample and pooling, our 2004–2010 reallocation term is 1.50 against their 1.48.

6 The cost of misallocation

The last exercise asks how much is at stake. If every firm in an industry earned the same return on its last worker and its last forint of capital, no reshuffling of inputs could raise output. The *efficiency gap* is how much more an industry would produce at that point with the same inputs, as a share of its value added; Hsieh and Klenow (2009) call it the potential gain. It is a model benchmark: adjustment costs, risk and measurement error also create differences in returns. Its change over time is more informative than its level.

6.1 Model and measurement

We follow Hsieh and Klenow (2009). Suppose every worker and every forint of capital earned the same return in every firm of an industry. The calculation asks how much more the industry would produce then, with the same total labour and capital. In the model each firm faces its own effective tax or subsidy, on its output or on its capital: a tax rate, an exemption, a regulation, a price cap, cheaper credit or a payment it must make to operate. Firms facing high effective taxes have high returns and are too small; favoured firms have low returns and are too large. What lowers aggregate output is how dispersed these effective rates are across firms, not their average level. The model assumes that firms in an industry sell differentiated products, and it weights industries by their value added.

Following Hsieh and Klenow (2009, Section IV), for firm i in two-digit industry s and year t, with value added PY_i, capital K_i, wage bill wL_i and capital share α_s:

$$
\begin{aligned}
TFPR_i &= \frac{PY_i}{K_i^{\alpha_s}(wL_i)^{1-\alpha_s}}, \qquad A_i \propto \frac{PY_i^{\sigma/(\sigma-1)}}{K_i^{\alpha_s}(wL_i)^{1-\alpha_s}}, \\
\overline{TFPR}_s &= \frac{\sum_i PY_i}{\left(\sum_i K_i\right)^{\alpha_s}\left(\sum_i wL_i\right)^{1-\alpha_s}}, \qquad A_s = \left(\sum_i A_i^{\sigma-1}\right)^{\frac{1}{\sigma-1}}, \\
\frac{TFP_s}{TFP_s^{\text{eff}}} &= \left[\sum_i \left(\frac{A_i}{A_s}\frac{\overline{TFPR}_s}{TFPR_i}\right)^{\sigma-1}\right]^{\frac{1}{\sigma-1}}, \\
\frac{Y^{\text{eff}}}{Y} &= \prod_s \left(\frac{TFP_s^{\text{eff}}}{TFP_s}\right)^{\theta_s},
\end{aligned}
$$

with θ_s the industry's share in value added. The efficiency gap is $100\,(Y^{\text{eff}}/Y - 1)$, computed year by year in nominal terms, $\sigma = 3$ as in Hsieh and Klenow. In the model a firm's $TFPR$ is high when it faces a tax-like distortion that keeps it small and low when it is favoured and too large. Misspecified technology, measurement error and demand assumptions also create $TFPR$ dispersion (Haltiwanger et al., 2018), so the efficiency gap is a frictionless-model benchmark, not a bound on the true gain in either direction.

The capital share is $\alpha_s = 1 - \text{median}\left[wL/(wL + D + r_t E)\right]$ over 5+ firm-years in the industry, 2004–2023 (median α_s 0.22), with D depreciation and E equity plus long-term debt; this is the same weight as in Section 5.1. Here, following Hsieh and Klenow, α_s is the capital share, one minus the labour weight denoted α in Sections 4 and 5. Hsieh and Klenow use US industry shares, not available at this level. Industries are two-digit (they use four-digit); coarser industries usually raise the measured gain, so levels are not comparable with theirs. Following them we trim the top and bottom 1% of $\ln(TFPR_i/\overline{TFPR}_s)$ and of $\ln(A_i M_s^{1/(\sigma-1)}/A_s)$ each year and recompute; industry-years with fewer than 10 firms are dropped. The firms with extreme measured productivity that are trimmed include many of the largest firms.

We use all market-economy firms with at least five employees, 2004–2023, and show manufacturing separately.

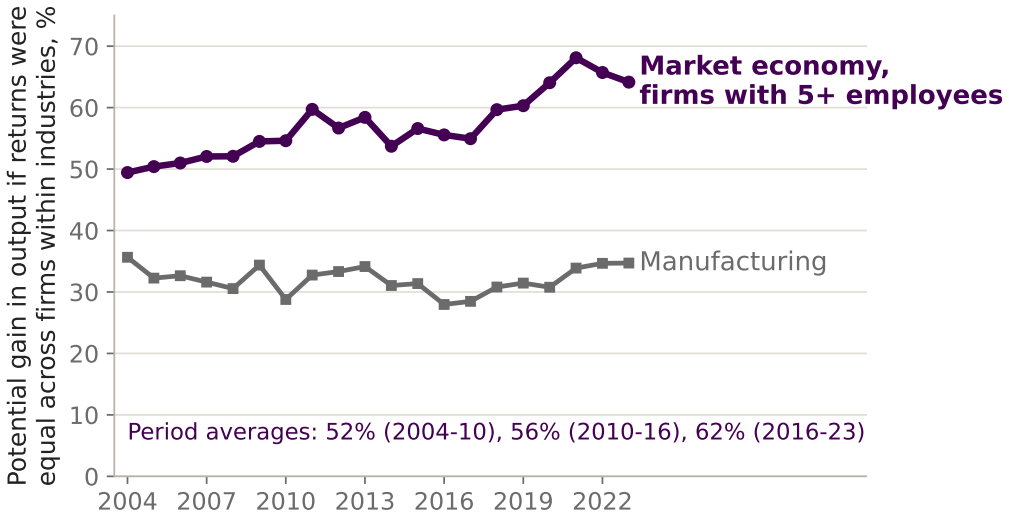


Figure 5: Efficiency gap: the potential gain in output from equalising returns within industries
Notes: Percentage increase in value added if marginal returns to labour and capital were equal across firms within each two-digit industry (Hsieh and Klenow, 2009), $\sigma = 3$, industries weighted by value-added shares, after trimming 1% of each tail. Market economy and manufacturing, firms with five or more employees. Sector detail in Figure 6.

Table 6: Efficiency gap, period averages

Efficiency gap, %	2004–2010	2010–2016	2016–2023
Market economy, 5+ employees (main)	52.0	56.5	61.6
CompNet industries, 20+ employees	42.2	45.9	46.2
Manufacturing, 5+ employees	32.3	31.3	31.6
Market economy, $\sigma = 5$	74.9	85.0	97.3
Market economy, capital cost without interest	43.0	45.3	50.0

Notes: Values of Figure 5 averaged over the years shown; the bottom rows vary the elasticity of substitution and the capital share.

6.2 Results

Figure 5 shows the efficiency gap. In the market economy (purple line) it was 52% of value added in 2004–2010; it rose by about six points by 2016–2019 and by about ten points including 2020–2023. In period averages it was 52% in 2004–2010, 56% in 2010–2016, 58% in 2016–2019 and 62% in 2016–2023 (peak 68% in 2021). Manufacturing (grey) did not change over the period as a whole: 31–32% throughout. Table 6 gives the period averages and how they vary with the elasticity of substitution and the capital share.

The rise is concentrated in construction, utilities, information and communication, accommodation and food, real estate and trade, and it happened within sectors (Figure 6). Gaps within sectors (log points, 2004–10 to 2016–23) rose in construction from 38 to 67, utilities 40 to 57, information and communication 30 to 42, accommodation and food 35 to 44, real estate 77 to 91 and trade 60 to 68; transport and administrative services fell slightly. Since $\ln(Y^{\text{eff}}/Y) = \sum_s \theta_s \ln(TFP_s^{\text{eff}}/TFP_s)$ is additive across industries, we can group industries into eleven sectors. Between 2004–2010 and 2016–2023 the aggregate rises from 41.9 to 47.9 log points: +6.4 from larger gaps within

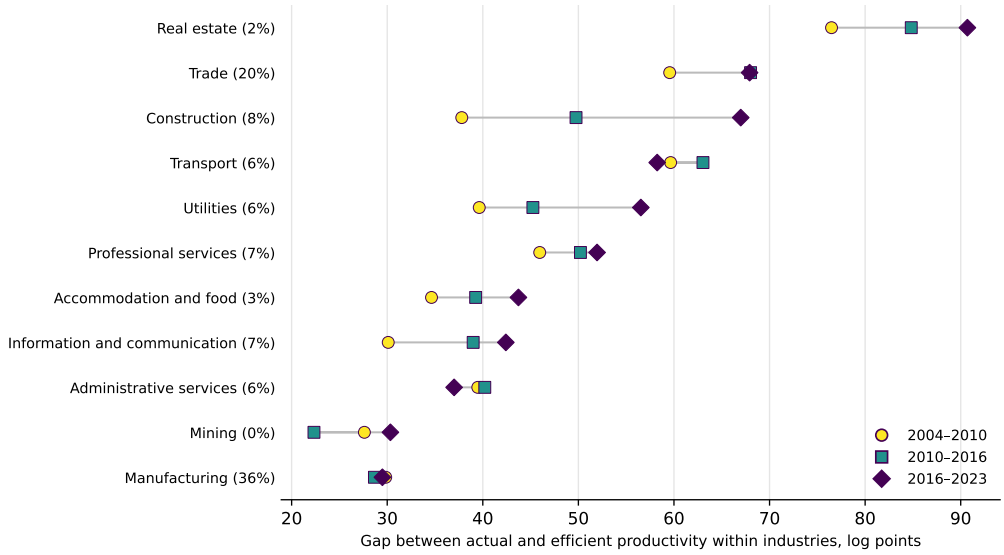


Figure 6: Efficiency gap by sector

Notes: Value-added-weighted average across the two-digit industries of each sector of $\ln(TFP^{\text{eff}}/TFP)$, in log points, main sample, period averages. Sector shares in value added in 2016–2023 in brackets.

sectors at initial shares and −0.4 from changes in shares (with symmetric weights at the two-digit level +6.8 and −0.8).

Among firms with 20 or more employees in CompNet's industries, which exclude utilities, the efficiency gap rose by four points, from 42% to 46%, and in manufacturing not at all (Table 6): the efficiency gap grew outside manufacturing, and more among firms with 5–19 employees than among larger ones. Among domestic firms the rise in the efficiency gap is at least as large as overall (Section 7).

6.3 Capital and labour

A more direct look separates the two inputs. For each firm we compare value added with its capital and with its wage bill, the returns on the two inputs, and measure how unequal these returns are across firms in the same industry. This needs no choice of weights, because the factor shares drop out. Between 2004–2010 and 2010–2016 the spread of returns on capital rose from 1.45 to 1.58 (standard deviation of logs) and then stayed there (1.57 in 2016–2023). The spread of returns on labour did not rise (0.55, 0.53 and 0.53). This is the pattern Gopinath et al. (2017) found in Spanish manufacturing, with similar trends in Italy and Portugal, and Gamberoni et al. (2016) in large euro-area economies: capital became more unevenly allocated, labour did not. The levels of the two spreads cannot be compared, because capital is costly to adjust and returns on it are more dispersed even without distortions (Asker et al., 2014). Only the change is informative, and part of it may reflect more volatile conditions rather than policy. The model itself does not attribute the efficiency gap to either input. Also, the return-on-capital dispersion rose early and then stayed high while the efficiency gap kept rising, so the rise in the gap also reflects how wedges line up with firm productivity, not only their spread.

6.4 Relation to the productivity gaps

The first and third exercises start from the same object: how unequal revenue productivity is across firms in an industry. Section 4 measures this spread and compares it across countries. This section puts it into a model that says how much output it costs. The two differ in three ways. First, the samples: Section 4 covers firms with 20 or more employees in CompNet's industries and weights industries by their number of firms; this section covers firms with five or more employees in the market economy and weights industries by value added. Second, the model's gain depends on how dispersed returns are and also on how the dispersion lines up with firms' underlying productivity. Third, the efficiency gap has no comparison with Poland.

The two agree where they overlap. Among firms with 20 or more employees in CompNet's industries, the sample of Section 4, the efficiency gap rose by only four points, and in the main sample the spread of revenue productivity itself hardly changed (standard deviation of log $TFPR$ within industry-years: 0.51, 0.52 and 0.52 in 2004–10, 2010–16 and 2016–23). This matches Figure 1, where the Hungarian gap stayed wide but did not widen. The larger rise in the efficiency gap comes from smaller firms, from industries outside manufacturing and from how the effective taxes line up with productivity. Hence the first exercise shows that the gaps did not close, and the third shows what such gaps cost and that this cost rose.

6.5 From the efficiency gap to the reserve

A useful benchmark is Hungary's own past. If the efficiency gap returned to its 2004–2010 average, the same workers and capital would produce 1–2% of GDP more. We call this the *reserve*. This is a one-off rise in the level of output, which then persists, not extra growth every year. If it took five years to reach, it would add 0.2–0.4 percentage points to growth a year over those five years.

The benchmark is $(1 + g_{\text{later}})/(1 + g_{2004\text{–}10}) - 1$: the proportional increase in output if the model's efficiency ratio returned to its 2004–2010 average, holding the later frontier fixed. In the main specification it is 6.3% of value added with 2016–2023 as the later period (6.2% from period averages of the log gain). It is an output-level difference, not an annual growth rate, and it does not restore the historical allocation firm by firm or model the transition.

To express it relative to GDP it has to be applied to a value-added base. The trim removes about 3% of firms but 31–44% of value added, depending on the year, because the top tail of the A_i deviation contains many of the largest firms. In 2023 the calculation keeps 60,935 of 62,857 firms and HUF 20,113 of 29,979 billion of value added. Applied to the firms kept, 6.3% is HUF 1,264 billion, 1.7% of 2023 GDP. Applied to all firms with 5+ employees, which assumes that the excluded firms would improve in the same proportion, it is HUF 1,884 billion, 2.5% of GDP.

The figure is sensitive to the model's choices and to the years compared, so we read it as a magnitude rather than a precise number. Table 7 shows the benchmark and its GDP share across variants. With 2016–2023 as the later period it ranges from 0.8% to 5.1% of GDP. It is above 1% of GDP in all variants except the 2% trim applied to the firms kept. With only the pre-pandemic years 2016–2019 as the later period it is smaller and more fragile, 0.2–2.0% of GDP, because the

Table 7: The return-to-2004–2010 benchmark across specifications, as a share of value added and of GDP

	2016–2023 vs 2004–2010			2016–2019 vs 2004–2010		
	Benchmark	% of GDP		Benchmark	% of GDP	
Variant	% of VA	kept firms	all 5+ firms	% of VA	kept firms	all 5+ firms
Trim 1% (main)	6.3	1.7	2.5	3.7	1.0	1.5
No trimming	6.7	2.6	2.7	0.6	0.2	0.3
Trim 0.5%	8.7	2.6	3.5	5.1	1.5	2.0
Trim 2%	3.6	0.8	1.4	1.5	0.4	0.6
$\sigma = 5$	12.8	–	5.1	–	–	–
Capital cost without financing cost	4.9	–	1.9	–	–	–

Notes: Benchmark: $(1 + g_{\text{later}})/(1 + g_{2004–10}) - 1$, from period averages of the annual gain, main sample unless stated. Share of 2023 GDP (HUF 75,218 billion), applied to the 2023 value added of the firms kept under each rule or of all market-economy firms with 5+ employees (HUF 29,979 billion). Trim: percentage at each tail of both deviations. $\sigma = 5$ and the capital-cost variant use the 1% trim; only the all-firm base is shown for them.

efficiency gap rose most in 2020–2023. The band of 1–2% covers the comparison with 2016–2019 in the main specification (1.0–1.5% of GDP) and the lower part of the comparison with 2016–2023 (1.7–2.5%), which includes the crisis years and the provisional 2023 data. The 2023 data matter less here than in the decomposition, because the model compares firms within the same year, but 2023 also sets the value-added base. Without trimming, a few extreme firms drive the level of the efficiency gap to about 200%, though the benchmark is similar (6.7%). For firms with 20+ employees in CompNet's industries the benchmark is 2.8% of their value added; in manufacturing alone it is about zero (−0.5%).

The level of the efficiency gap depends on the model (CES demand with a common σ, Cobb–Douglas technology with industry factor shares), on two-digit industries and on trimming. Hungarian factor shares may themselves contain distortions. Capital is measured at book value. The benchmark does not identify which policies created the wedges, or what a particular reform would deliver. The GDP share compares a value-added change for market-economy firms with 5+ employees to total GDP, so it ignores the rest of the economy. For these reasons we report a magnitude, a one-off rise in the level of output of 1–2% of GDP (0.2–0.4 percentage points of growth a year if reached over five years), not a point estimate.

6.6 Poland

Gradzewicz (2026) applies the method to Polish firms with 10+ employees and finds that the efficiency gap in Poland also rose, from close to 40% of value added in the 1990s to 70–75% in the early 2020s. This measure therefore does not show the contrast between Hungary and Poland of Section 4, and the rise over time cannot be read as specific to Hungary. The two Polish results, narrowing gaps and a rising efficiency gap, use different samples and definitions, and we cannot reconcile them; this is one more reason to read the efficiency gap as a magnitude. The studies differ in samples and assumptions, so their levels are not comparable.

7 Foreign-owned and domestic firms

Foreign-owned firms employ 37–39% of the workers in our sample of firms with five or more employees. Is weak allocation a problem of domestic firms, often small ones, or does it affect foreign-owned firms too? Table 8 splits both the contribution of resources moving (Section 5) and the efficiency gap (Section 6) by ownership.

Table 8: Foreign-owned and domestic firms: resources moving and the efficiency gap

A. Exercise 2: contribution of resources moving, percentage points a year	2004–13	2013–22	Change
All firms	1.58	0.84	−0.75
of which foreign-owned firms	0.62	0.15	−0.47
of which domestic firms	0.96	0.68	−0.28
Domestic firms only, separate decomposition	1.50	1.12	−0.38
Employment share of foreign-owned firms, %	37	39	
B. Exercise 3: efficiency gap, % of value added	2004–10	2016–19	2016–23
All firms	52	58	61
Domestic firms	49	58	61

Notes: Panel A: reallocation plus net entry and exit as in Table 4, averages of three three-year windows. The foreign and domestic parts add up to the value for all firms. Domestic firms only: the decomposition redone without foreign-owned firms. Employment share of foreign-owned firms at the start of each window. Ownership is measured at the start of each window, for entrants at its end. Panel B: efficiency gap as in Figure 5, period averages, firms with 5+ employees; both rows cover the industries of the domestic sample and weight them by their value-added shares in the full sample.

Because each term of the decomposition is a sum over firms, the contribution of resources moving splits exactly into a part that comes from foreign-owned firms and a part that comes from domestic firms (Panel A). Before 2013 foreign-owned firms contributed 0.6 points a year and domestic firms 1.0; after 2013, 0.15 and 0.7. The contribution of foreign-owned firms fell by three-quarters, that of domestic firms by under a third. About 60% of the fall (0.47 of 0.75 points) therefore comes from foreign-owned firms, more than their share of employment, and their contribution after 2013 is below its 2004–2007 and 2010–2013 levels in every window. Redoing the decomposition on domestic firms only, the contribution fell from 1.5 to 1.1 points a year, by about a quarter.

Panel B compares the efficiency gap. With the same industries and industry weights, the efficiency gap among domestic firms rose from 49% of their value added in 2004–2010 to 58% in 2016–2019 and 61% in 2016–2023; for all firms it was 52%, 58% and 61%. Domestic firms start from a slightly smaller gap and end at the same level, so the rise among them is at least as large as overall. With the domestic firms' own industry weights the rise looks larger (49% to 64%), but that reflects where domestic value added is concentrated, not a difference within industries.

So allocation weakened in both groups, in different ways. Among foreign-owned firms the flow nearly stopped (0.15 points a year after 2013), and Koren and Rátfai (2026) find that their employment growth advantage over domestic firms that do not export has disappeared. Part of this is the end of the expansion of foreign-owned firms before the crisis (Muraközy et al., 2018). Part of it may come from policy, because foreign-owned firms were on both sides of firm-specific

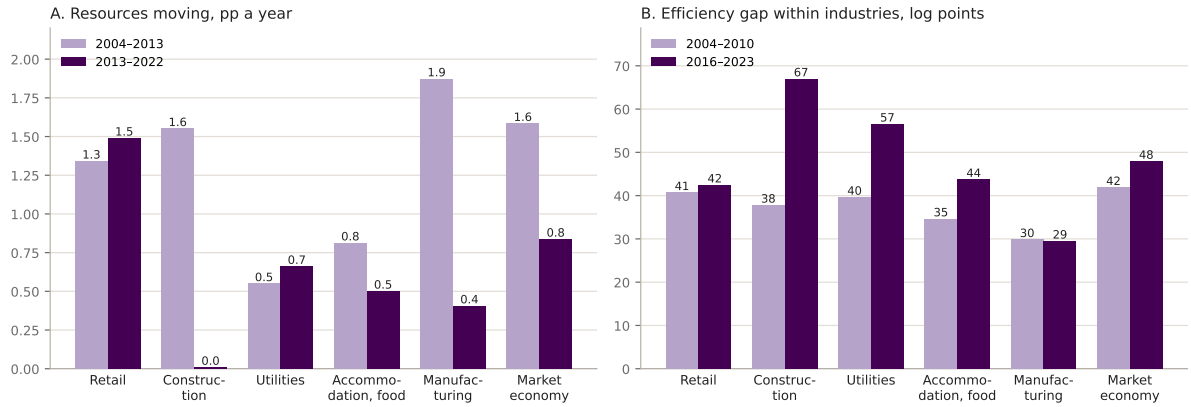


Figure 7: Four industries: contribution of resources moving and the model's efficiency gap

Notes: Panel A: reallocation plus net entry and exit (Section 5, main specification), annual percentage points, two-digit industries weighted by their employment, averages of the three-year windows before and after 2013. Panel B: $\ln(TFP^{\text{eff}}/TFP)$ within two-digit industries (Section 6, main specification), log points, weighted by value added, period averages. Retail: NACE 47; construction: 41–43; utilities: 35–39; accommodation and food: 55–56; manufacturing: 10–33. Firms with five or more employees.

measures. The sector-specific special taxes introduced after 2010, in retail, telecommunications, energy and banking among others, were designed in a way that allowed discrimination against foreign-owned taxpayers (Varju and Papp, 2022). At the same time, since 2012 the government has signed strategic cooperation agreements with 100 large companies, nine in ten of them foreign-owned (HIPA Hungarian Investment Promotion Agency, 2025). Both kinds of measure make how a firm is treated depend on who it is rather than on general rules. Among domestic firms the flow slowed less, but the efficiency gap widened at least as much as overall. Our data do not show which measures, if any, caused these changes.

For policy, this suggests that equal treatment matters for foreign-owned firms as much as for domestic ones: special taxes that single out some firms, and agreements or subsidies that favour others, both make effective tax rates differ across firms in the same industry. The ownership data change source in 2015 and again after 2020, so we read the split as indicative.

8 Four industries

Policy makers want to know where the problem is. Figure 7 shows the decomposition and the efficiency gap for four industries with heavy sector-specific intervention since 2010, such as special taxes, regulated prices, price caps and public procurement: retail, construction, utilities, and accommodation and food. They were chosen before the results were computed. Manufacturing and the market economy are shown for comparison.

Figure 7 compares the earlier period (light bars) with the later one (dark bars). Panel A is the contribution of resources moving to productivity growth (higher is better); Panel B is the model's efficiency gap (higher is worse). The efficiency gap rose most in construction (38 to 67 log points) and utilities (40 to 57), less in accommodation and food (35 to 44), and hardly at all in retail (41 to 42) and manufacturing (30 to 29). The contribution of resources moving is noisier at the

Table 9: What each exercise measures, and what it does not tell us

	What it measures	What it does not tell us
Gaps (Section 4)	How far productivity differs between firms in the same industry, compared with Poland and the region (firms with 20+ employees; Poland and the region to 2019)	Whether the gaps come from policy, from prices and markups, or from measurement
Reallocation (Section 5)	How much of productivity growth came from workers moving between firms and from firms entering and leaving the sample (5+ employees)	Why workers moved less, or how capital moved
Efficiency gap (Section 6)	What a standard model says the same inputs would produce if returns were equal across firms, and how this changed	The effect of any specific policy change

level of an industry, because the three-year windows swing widely, sometimes from positive to negative. Two patterns are clear. In construction it fell from 1.6 points a year to about zero, and in manufacturing from 1.9 to 0.4. In retail it did not fall (1.3 and 1.5),[5] and in utilities and in accommodation and food the windows vary too much to read a trend.

Construction is the industry where both measures deteriorated. Manufacturing shows the other combination: resources moved much less after 2013, but the efficiency gap, measured within the year, did not widen.

9 Discussion

9.1 Putting it together

Table 9 summarises what each exercise measures and what it does not tell us.

The decomposition measures a flow: how much productivity growth comes from workers and capital moving between firms. The productivity gaps and the efficiency gap measure a state: how far apart firms are, and how far allocation is from efficient, in a given year. The two can move differently, and they did.

The flow weakened after 2013, most among foreign-owned firms; among domestic firms it slowed by about a quarter. The state deteriorated. The model's efficiency gap rose, among domestic firms at least as much as in the economy as a whole, most in construction and utilities, and returns on capital became more uneven across firms. The productivity gaps show that the reallocation that did happen was not enough to bring Hungary's gaps down to Poland's level. We do not observe Poland's reallocation flow, so we cannot say whether resources moved faster there or whether less new dispersion was created.

[5] Muraközy et al. (2018, Chapter 8) found weaker reallocation in retail after 2010 than before the crisis. With our split at 2013 we see no further decline after 2013. Section 9.2 returns to retail.

Together the exercises suggest that workers moved less towards more productive firms after 2013, that returns on capital became more uneven across firms, and that the efficiency gap grew.

9.2 Policy implications

The question we started from is whether there are resources in the Hungarian economy that better policy could mobilise without new spending. Our results suggest that there are. Hungary has productive firms; the problem is that they do not grow enough at the expense of less productive ones. In an economy with gaps as wide as Hungary's, workers and capital would produce more in the firms that use them best. The benchmark of Section 6.5 gives the order of magnitude at stake if allocation merely returned to where it was fifteen years ago. Reaching it takes time, because moving resources has adjustment costs, and it depends on the policy environment more than on any single measure.

Our evidence does not identify which policy slowed reallocation. The literature says which kinds of policy can. Restuccia and Rogerson (2008) showed that taxes and rules that treat firms differently by identity rather than by general criteria lower aggregate productivity even when no firm becomes less efficient. Andrews and Cingano (2014) linked allocative efficiency to product market regulation, employment protection and barriers to entry. Andrews et al. (2017) showed that weak firms that survive on soft financing crowd out productive ones.

In the language of the efficiency gap, anything that makes the effective tax rate differ across firms in the same industry adds to the loss: taxes and exemptions tailored to particular firms or groups, discretionary subsidies and procurement, and corruption, which works like a tax that some firms pay and others do not. With firm-level data from Uganda, Fisman and Svensson (2007) found that a one percentage point higher bribery rate was associated with about three percentage points lower firm growth, roughly three times the effect of taxation.

Retail trade is a Hungarian example that has been studied in detail. Muraközy et al. (2018, Chapter 8) found that before the 2008 crisis productivity growth in retail came mainly from reallocation, chiefly through entry and the expansion of large chains, many of them foreign owned. This stopped after 2010, and the contribution of reallocation fell; they suggest that this may be related to the size-dependent measures introduced after 2010, such as the crisis taxes on retail.

The Hungarian pattern, with gaps between firms wider than in Poland on every measure (Section 4.4), returns on capital becoming more uneven across firms, and an efficiency gap that grew most in construction and utilities but not in exposed manufacturing (Sections 6 and 8), is consistent with these mechanisms: an environment in which the growth of firms depends on factors other than their productivity.

Three principles follow, none requiring public money. First, rules should apply equally to all firms and be stable, so that firms grow because they are productive, not because they are favoured. This holds for foreign-owned firms too: some sector-specific taxes allowed discrimination against them, while nine in ten strategic agreements with the government were signed with foreign-owned companies (Section 7). Second, measures that change margins in particular industries, such as

price caps and sector-specific taxes, should be temporary by design and reviewed against their effect on allocation. Third, entry and exit should be open: in our data most of the contribution of entry and exit comes from weak firms leaving, and this contribution fell from 1.3 percentage points a year before 2013 to 0.6 after. The literature on weak firms kept alive by soft financing (Andrews et al., 2017) describes one way this can happen; our data do not show whether it did.

A last point concerns the composition of growth. Much of economic policy aims at growing the capital stock, including through support for firms' investment. Reallocation works through a different channel: it raises total factor productivity, the output obtained from given inputs. Hence growth from better allocation can continue even if support aimed at growing the capital stock is reduced. If capital and labour flow to the firms that use them best, the economy can grow faster on the same, or a smaller, investment budget. Such gains come gradually, so they should be tracked. The gaps between firms, the contribution of resources moving and the efficiency gap can be updated every year from firm balance sheets; the central bank's productivity report does not cover them.

9.3 Caveats

Finally, let us point out the limits of our evidence. We document how allocation changed, but we cannot tell which policy, if any, caused the change. Each exercise also leaves questions open, as Table 9 shows. Gaps in revenue productivity can reflect prices, markups and measurement as well as misallocation, and the model behind the efficiency gap is a benchmark, not an estimate of what a reform would deliver.

Some results are robust: the gap to Poland and the old member states in the late 2010s, the fall in the contribution of resources moving between 2004–2013 and 2013–2022 in every specification we tried, and the sectoral pattern of the model's efficiency gap. The level of the efficiency gap is large in every specification, but it remains a model benchmark rather than an estimate of the output lost.

Other results are less robust. The Hungary–Poland *trend* contrast depends on the measure, and so does any split between labour and capital (Section 4.4). The decline in reallocation is volatile from window to window, and its size differs across methods (it is larger with the Melitz–Polanec decomposition). A similar rise in the efficiency gap is reported for Poland. The size of the reserve depends on the model's choices and on the period compared. Finally, the ownership data change source in 2015 and after 2020, so the split by ownership is indicative.

Several results lean on 2020–2023: the weak reallocation window of 2019–2022 and much of the rise in the efficiency gap. These were the years of the pandemic, the energy shock and the price caps, and the 2023 data are provisional. Data for 2024 will be added soon. They will help to tell whether the weakening persists once the crisis years pass or whether, as after 2013–2016, a recovery follows.

10 Conclusion

This paper asked whether part of the growth Hungary needs can come from workers and capital moving from less to more productive firms. Using balance sheets of all Hungarian firms for 2004–2023, we find that productivity gaps between firms stayed wider than in Poland and the region, that the contribution of resources moving to productivity growth fell by almost half after 2013, and that the model's efficiency gap rose, among domestic firms at least as much as overall. Returning to the allocation of 2004–2010 would raise the level of output by 1–2% of GDP.

Our results are consistent with resources, capital in particular, following productivity less than they used to among domestic firms, and with the cost of this having risen, but they do not prove it. Linking these changes to specific policies is the next step. Policy that gives firms more freedom could mobilise this reserve without new public spending.

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A Supplementary figures

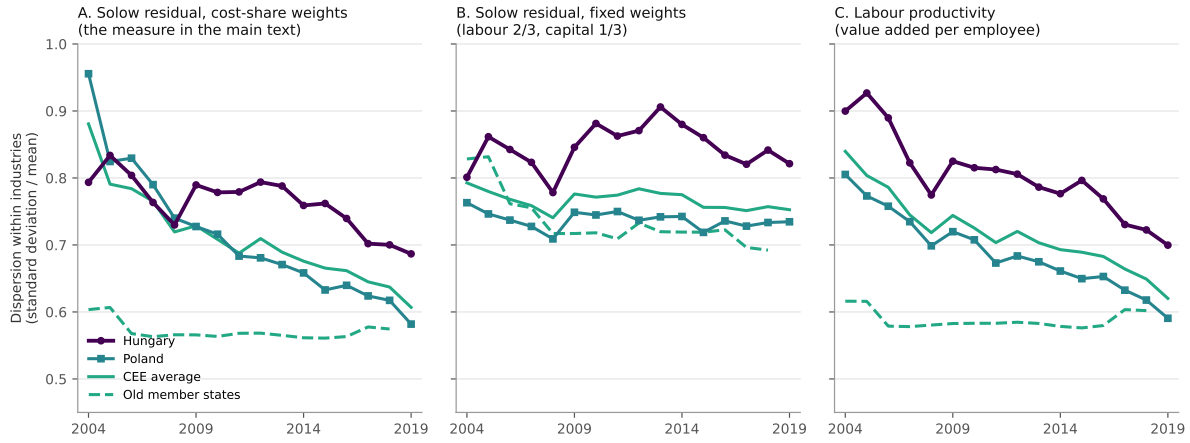


Figure A1: Three CompNet measures of productivity gaps, 2004–2019

Notes: As Table 2, yearly values. Hungary is CompNet's own line.

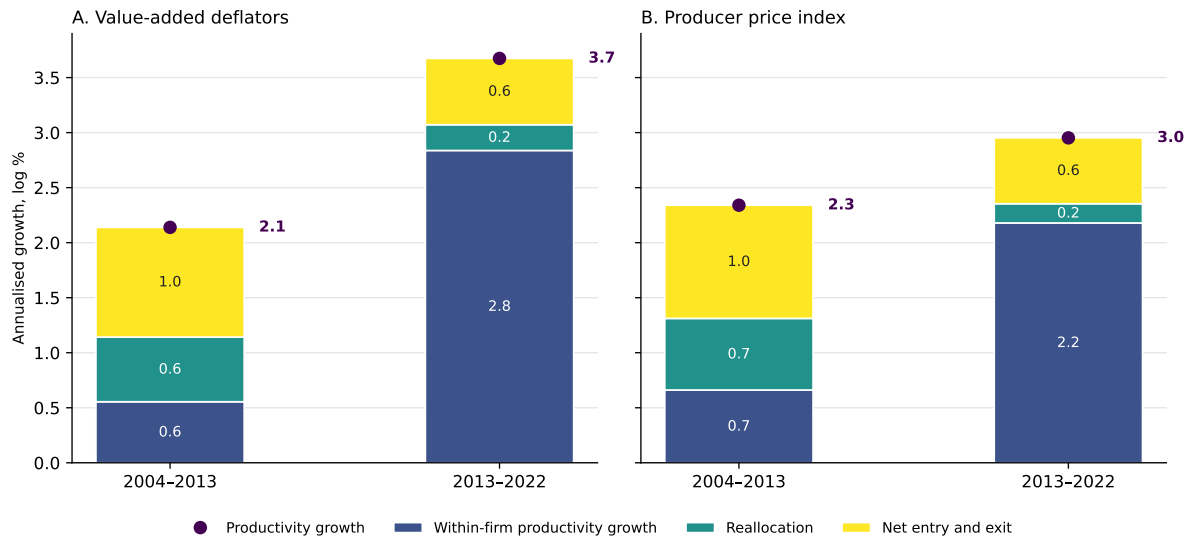


Figure A2: All components of productivity growth before and after 2013, two price indices

Notes: As Table 4, 2004–2013 and 2013–2022. Panel A: industry value-added price indices from the national accounts (Eurostat, 2026). Panel B: industry producer price indices.

B Replication and checks

The results come from three make-based pipelines (Stata 18 and Python 3) in folders `tfp/misallocation`, `tfp/reallocation` and `tfp/hsieh-klenow`, each with a README, a decision log and aggregated outputs. All tables are built from output files by one script; numbers quoted in the prose were checked against those files but are typed. The dispersion series and the Hsieh–Klenow efficiency gaps were re-implemented independently in Python (within 0.0003 and to machine precision). Two independent reviews of earlier versions and the responses to them are filed with

the code. Firm-level data are licensed and stay in the research environment; only aggregates are published.

This is a draft. Work in progress includes: (i) regressions of firms' employment and capital growth on their earlier productivity, by period, to test directly whether resources respond less to productivity than before; (ii) capital-only and labour-only versions of the Hsieh–Klenow calculation, to test inside the model whether the rise in the efficiency gap comes from capital; (iii) a sample bridge showing firms, employment, value added and capital through each filter, and dispersion with frozen factor weights on Hungarian data; (iv) an update of the regional comparison with the 10th vintage of CompNet; (v) generating all numbers in the text directly from the output files.